Platinum-loaded TiO2 for photocatalytic hydrogen evolution

We report a simple impregnation route to Pt-loaded TiO2 nanoparticles. The material shows a band gap of 3.2 eV and a hydrogen evolution rate of 450 umol/g/h under UV light. Stability was tested over five cycles of 4 h each without loss in activity.

Sample	Band gap	H2 yield
TiO2	3.3 eV	12 umol/g/h
TiO2-Pt	3.2 eV	450 umol/g/h

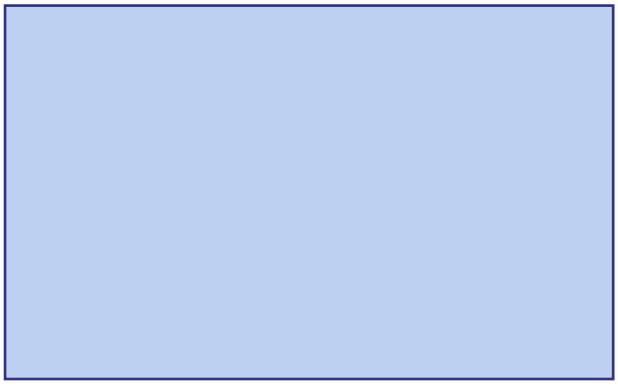

Figure 1: H2 evolution over time.

Conclusion: Pt loading increases activity by a factor of 37.

Code: github.com/JSR-ISM-Smart-Chemistry-Lab/Titanarium